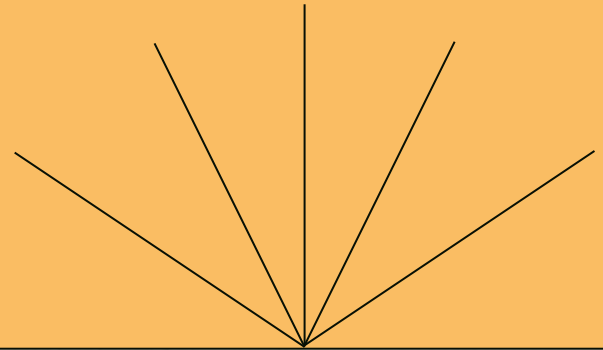
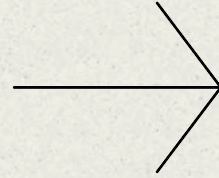


Company Project Management System Upgrade

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BUAD 346



Statement of Problem:

Situation: My uncle's company (PELSA Company) needs to update the project database storage.

Current structure:

- Exclusively Google Sheets
- 2000-10,000 projects stored per sheet, 12+ sheets of information
- Employees and tech admin have same access and editing abilities in the database

Goals with the new data management system:

- Easily buildable and sustainable
 - Can have backend coding, but ideally minimal coding for upkeep
- Accessible by different users (Admin and Employees)
- Simple UI for employees to create, search, and update records

New Database Options:

- PostgreSQL+ with NocoDB
- Grist
- AirTable

Gut Instinct:

- PostgreSQL+ with NocoDB due to its customizable nature.

Proposed Methods of Comparison and Decision Making



Scoring
Model

AHP

Scoring Model

- Simple and fast way to compare multiple platform options
- Easy to assign weights to the criteria that matter most
- Pretty simple to understand at face value, which is good for stakeholders
- Very Appropriate for this method
 - Involves lots of varied criteria
 - We already know our priorities
- Assumes that higher score is the better solution and that weights reflect actual priorities. Additionally, it assumes that all the criteria are independently related and that there is one standard agreed upon ranking.

Advantages

- Simple, fast, and easy to understand while still allowing multiple criteria to be compared at one time.

Disadvantages

- Relies heavily on subjective ratings and does not check consistency, which can lead to inaccurate or biased information.

AHP

- Provides more depth and structure to the decision process
- Gives more accurate weight representation
- Has flexibility for more complex decisions with multiple criterias
- Definitely appropriate to use in this situation
 - With so many criteria, it needs to be accounted for
 - Has better capability to see if it would be beneficial for long term implementation
- Assumes that criteria are independent and that these different pieces can be represented one dimensionally. Preferences are consistent and biases don't affect it.

Advantages

- More structured and accurate because it uses the comparisons and tests the consistency to make the decisions.

Disadvantages

- More time-consuming and complex due to the large number of comparisons required, harder to understand at face value.

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Analysis and Recommendation

Scoring Model

			\$24 a month/seat for NocoD		\$24 a month/user		\$20 a month/seat		
Scoring Model			PostgreSQL + NocoDB		Grist		Airtable		
Criteria	Weight		Rating	Score	Rating	Score	Rating	Score	
Pricing (Low)	3		5	15	5	15	7	21	
Setup Effort (Minimal)	4		4	16	6	24	7	28	
UI Simplicity	5		6	30	8	40	8	40	
Ease of Maintenance	3		6	18	8	24	7	21	
Permissions	4		9	36	6	24	7	28	
Adapting to New Employee View	5		7	35	9	45	7	35	
Customization	3		10	30	7	21	5	15	
Sum				180		193		188	
Percent of Highest	264			25		41.28787879		33.71212121	

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Pricing					Averages			
	PoSQL + NocoDE	Grist	Airtable	PoSQL + NocoDE	Grist	Airtable	Average	
PoSQL + NocoDB	1.00	1.00	0.71	0.29	0.29	0.29	0.29	0.29
Grist	1.00	1.00	0.71	0.29	0.29	0.29	0.29	0.29
Airtable	1.40	1.40	1.00	0.41	0.41	0.41	0.41	0.41
Sum	3.40	3.40	2.43	1.00	1.00	1.00	1.00	
Set up Effort					Averages			
	PoSQL + NocoDE	Grist	Airtable	PoSQL + NocoDE	Grist	Airtable	Average	
PoSQL + NocoDB	1.00	0.67	0.57	0.24	0.24	0.24	0.24	0.24
Grist	1.50	1.00	0.86	0.35	0.35	0.35	0.35	0.35
Airtable	1.75	1.17	1.00	0.41	0.41	0.41	0.41	0.41
Sum	4.25	2.83	2.43	1.00	1.00	1.00	1.00	
UI Simplicity					Averages			
	PoSQL + NocoDE	Grist	Airtable	PoSQL + NocoDE	Grist	Airtable	Average	
PoSQL + NocoDB	1.00	0.75	0.75	0.27	0.27	0.27	0.27	0.27
Grist	1.33	1.00	1.00	0.36	0.36	0.36	0.36	0.36
Airtable	1.33	1.00	1.00	0.36	0.36	0.36	0.36	0.36
Sum	3.67	2.75	2.75	1.00	1.00	1.00	1.00	
Ease of Maintenance					Averages			
	PoSQL + NocoDE	Grist	Airtable	PoSQL + NocoDE	Grist	Airtable	Average	
PoSQL + NocoDB	1.00	0.75	0.86	0.29	0.29	0.29	0.29	0.29
Grist	1.33	1.00	1.14	0.38	0.38	0.38	0.38	0.38
Airtable	1.17	0.88	1.00	0.33	0.33	0.33	0.33	0.33
Sum	3.50	2.63	3.00	1.00	1.00	1.00	1.00	

Permissions					Averages			
	PoSQL + NocoDE	Grist	Airtable	PoSQL + NocoDE	Grist	Airtable	Average	
PoSQL + NocoDB	1.00	1.50	1.33	0.41	0.41	0.42	0.41	
Grist	0.67	1.00	0.86	0.28	0.27	0.27	0.27	
Airtable	0.75	1.17	1.00	0.31	0.32	0.31	0.31	
Sum	2.42	3.67	3.19	1.00	1.00	1.00	1.00	
Adapting to New Employee View					Averages			
	PoSQL + NocoDE	Grist	Airtable	PoSQL + NocoDE	Grist	Airtable	Average	
PoSQL + NocoDB	1.00	0.75	1.00	0.30	0.30	0.30	0.30	
Grist	1.33	1.00	1.33	0.40	0.40	0.40	0.40	
Airtable	1.00	0.75	1.00	0.30	0.30	0.30	0.30	
Sum	3.33	2.50	3.33	1.00	1.00	1.00	1.00	
Customization					Averages			
	PoSQL + NocoDE	Grist	Airtable	PoSQL + NocoDE	Grist	Airtable	Average	
PoSQL + NocoDB	1.00	1.43	2.00	0.45	0.45	0.45	0.45	
Grist	0.70	1.00	1.40	0.32	0.32	0.32	0.32	
Airtable	0.50	0.71	1.00	0.23	0.23	0.23	0.23	
Sum	2.20	3.14	4.40	1.00	1.00	1.00	1.00	

AHP

	Pricing (Low)	Setup Effort (Minimal)	UI Simplicity	Ease of Maintenance	Permissions	Adapting to New Employee View	Customization
PoSQL + NocoDB	0.294117647	0.235294118	0.272727273	0.285714286	0.413598153	0.3	0.454545455
Grist	0.294117647	0.352941176	0.363636364	0.380952381	0.272415353	0.4	0.318181818
Airtable	0.411764706	0.411764706	0.363636364	0.333333333	0.313986494	0.3	0.227272727
	Relative Priority			Total Rankings			
Pricing (Low)	0.032984091		PoSQL + NocoDB	0.315286352			
Setup Effort (Minimal)	0.134136243		Grist	0.358788593			
UI Simplicity	0.196324724		Airtable	0.325925055			
Ease of Maintenance	0.062557132						
Permissions	0.103460061						
Adapting to New Employee View	0.349829625						
Customization	0.120708124						

Adjustments to Scoring Model

		\$24 a month/seat for NocoDB		\$24 a month/user		\$20 a month/seat	
Scoring Model		PostgreSQL + NocoDB		Grist		Airtable	
Criteria	Weight	Rating	Score	Rating	Score	Rating	Score
Pricing (Low)	4	4	16	4	16	5	20
Setup Effort (Minimal)	4	4	16	4	16	7	28
UI Simplicity	5	6	30	8	40	8	40
Ease of Maintenance	4	6	24	6	24	7	28
Permissions	5	9	45	6	30	7	35
Adapting to New Employee View	5	7	35	8	40	7	35
Customization	3	10	30	7	21	5	15
Sum			196		187		201
Percent of Highest	271		27.7		29.5202952		42.804428

Issues that could arise:

- If how pricey the options are becomes more of an issue
- If the set up of Grist was more difficult
- If it was harder to maintain Grist than anticipated

Recommendation: Grist

- Scored highest by 6 points in the scoring model
- Ranked highest at AHP
- It hits the balance between customizable and functional while being accessible to non technical users
- Maintenance, scalability, and user experience will grow with the company smoothly
- Has the flexibility and buildability of PostgreSQL+ with NocoDB but the user friendliness and approachability of Airtable.

Even though it went against the gut solution, it seems to be the most realistic and best option to implement in the business. I will continue to do further research and will likely implement this as the company project database management system